GIS Wetland Degradation Assessment

A spatial assessment of wetland-associated vegetation condition, spectral change and surface-water evidence between two observation dates.

OBSERVATION WINDOW	2023-08-14 → 2023-08-28	INTERVAL	14 days
WETLAND CONTEXT	Project-provided wetland boundary	ANALYSIS BASIS	RASTER_CRS_AND_TRANSFORM

OVERALL FINDING

Measurable ecological change was detected in the wetland analysis area.

The comparison contains measurable vegetation change. Its significance is evaluated below together with surface-water evidence rather than being treated as physical wetland loss by itself. Surface-water change reports 0.00 ha of loss and 3.78 ha of gain. Physical wetland depletion is not established by this run. The engine explicitly prevents vegetation decline or optical water change from being presented as physical wetland loss without independent supporting evidence.

VEGETATION DEGRADATION	0 px · 0.00%
VEGETATION RECOVERY	0 px · 0.00%
STABLE CLASSIFIED STATE	3,500 px · 100.00%
MEAN CVA MAGNITUDE	0.6969

Interpretation boundary: the finding describes change detected by the available remote-sensing evidence. It does not by itself establish that the wetland is ecologically healthy or that its physical extent has remained unchanged.

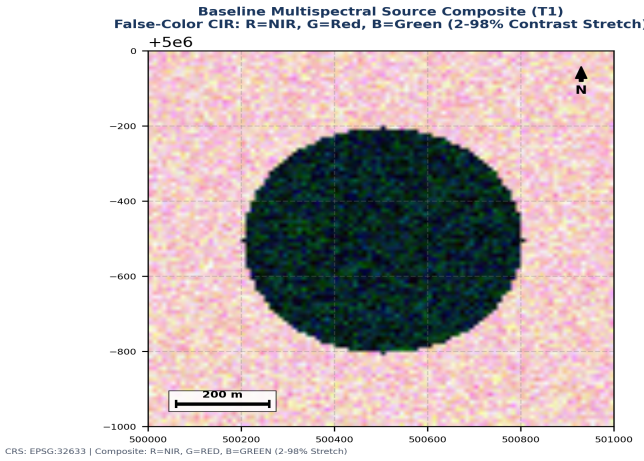
01 WETLAND UNDER ASSESSMENT

The object of this assessment is the **wetland analysis area** defined by the wetland-context stage of the GIS engine. This spatial context is then used to interpret vegetation and hydrological observations. The wetland is therefore not being defined simply as “land with moisture”; it is a spatial analytical unit supplied or inferred by the configured wetland-mask method.

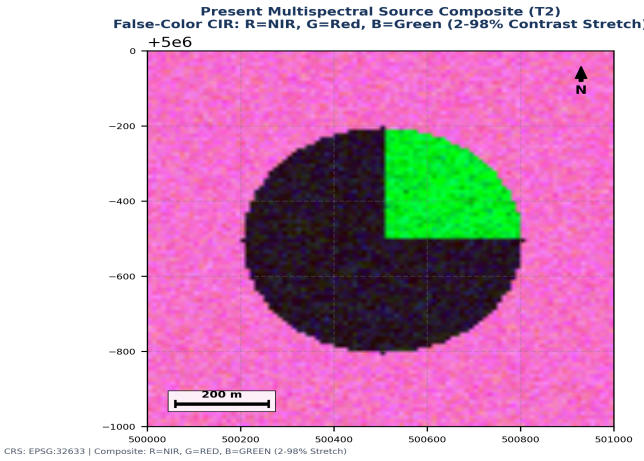
The analysis is spatially anchored to the supplied wetland boundary/data source. Vegetation and hydrological observations are interpreted within that wetland context.

FIELD	OBSERVED / CONFIGURED VALUE
Wetland context status	WETLAND_BOUNDARY_AVAILABLE
Wetland source	Project-provided wetland boundary
Source dataset / boundary	wetland_boundary.geojson
Wetland area	36.00 ha
Valid wetland analysis area	24.00 ha
Physical depletion pathway	NOT_ASSESSED

SPATIAL CONTEXT — SOURCE IMAGERY



T1 — baseline source composite (2023-08-14)



T2 — present source composite (2023-08-28)

These are the actual multispectral source composites used by the report cartography. They provide the landscape context for the wetland analysis; the wetland mask itself remains an analytical classification layer and is not presented as a solid standalone image.

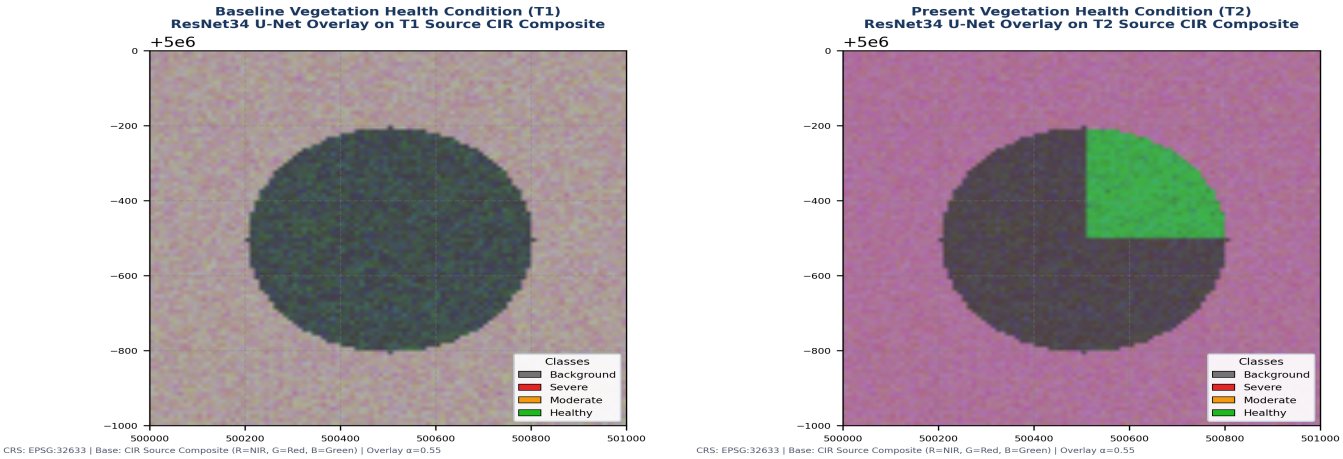
What the wetland result means

A wetland mask answers *where the wetland is considered to be for this analysis*. It does not, on its own, answer whether the wetland is degraded. Degradation is evaluated from temporal evidence inside that analytical context.

02 VEGETATION CONDITION WITHIN THE WETLAND

Vegetation condition is inferred independently at T1 and T2 using the configured ResNet34 U-Net classification. The classes describe vegetation condition, not wetland type: Healthy Vegetation, Moderate Stress, Severe Stress, and Background / Non-Vegetated.

CONDITION CLASS	T1 PIXELS	T1 %	T2 PIXELS	T2 %	CHANGE
Healthy vegetation	0	0.00%	0	0.00%	+0 px
Moderate stress	0	0.00%	0	0.00%	+0 px
Severe stress	0	0.00%	0	0.00%	+0 px
Background / non-vegetated	10,000	100.00%	10,000	100.00%	+0 px



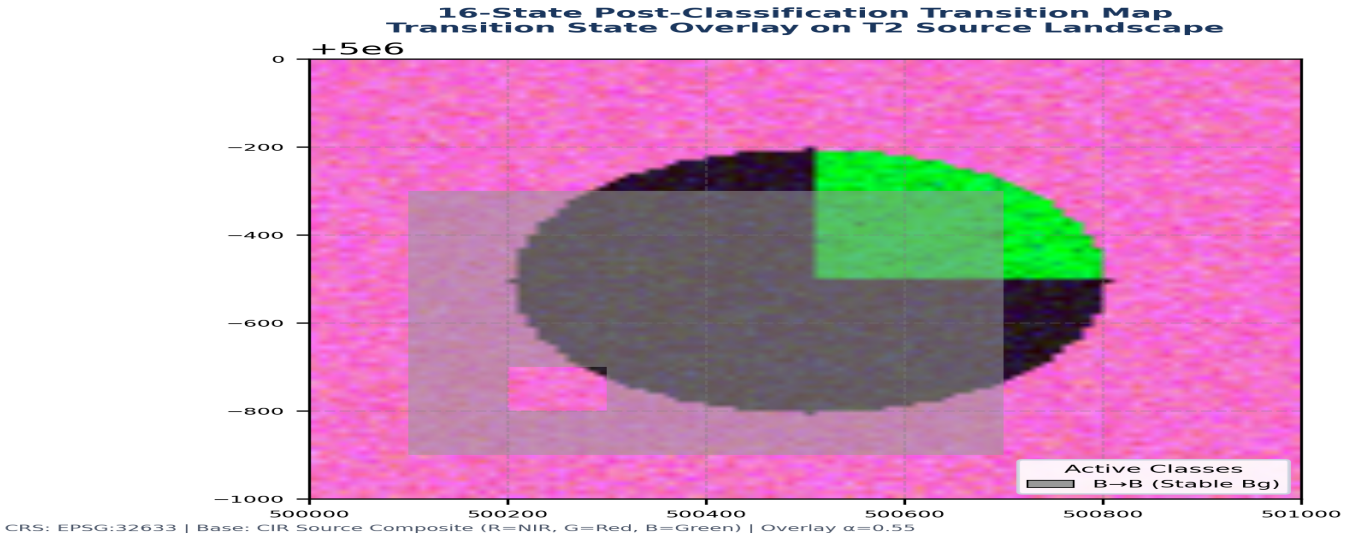
Left: baseline vegetation condition (2023-08-14). Right: present vegetation condition (2023-08-28).

Interpretation

The classified vegetation state is unchanged at the aggregate level: Healthy Vegetation remains at 0 pixels, while the degradation transition count is 0. This is evidence of **no detected vegetation condition deterioration** over the comparison interval, not proof of long-term wetland health.

03 VEGETATION CHANGE PATHWAYS

The transition map records the state occupied by each analyzed pixel at T1 and T2. This is more informative than comparing two totals because it shows whether vegetation moved toward stress, recovered, disappeared into background, or remained in the same class.



PATHWAY	DEFINED TRANSITIONS	PIXELS	PROPORTION
Persistent classified state	H→H, M→M, S→S, B→B	3,500	100.00%
Movement toward poorer vegetation condition	H→M, H→S, M→S	0	0.00%
Movement toward better vegetation condition	M→H, S→M, S→H	0	0.00%
Canopy loss / non-vegetated transition	H→B, M→B, S→B	0	0.00%
Emergence; context dependent	B→H, B→M, B→S	0	0.00%

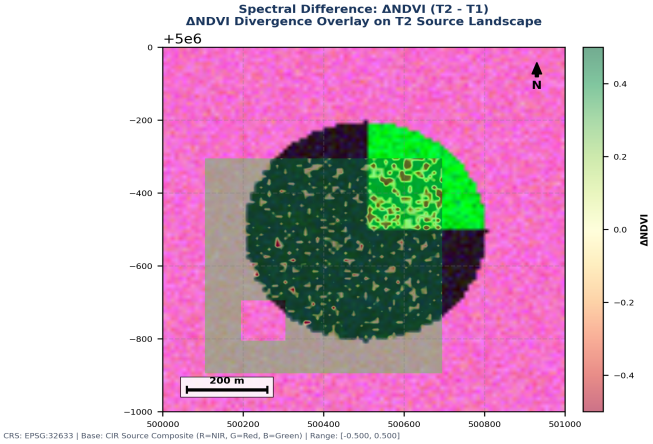
Interpretation

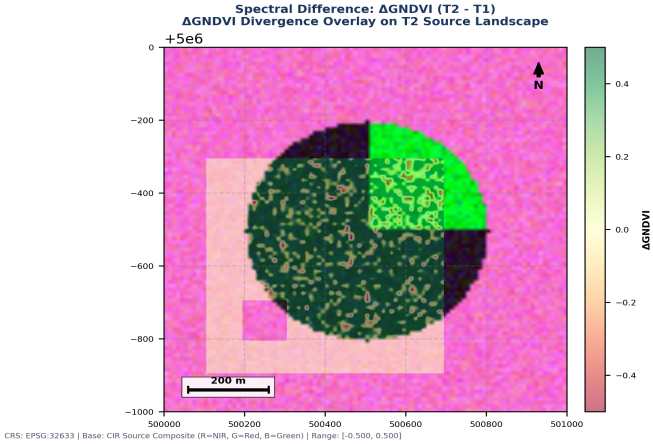
No pixels followed the engine's defined vegetation-degradation pathways (H→M, H→S, M→S), and no pixels followed the defined recovery pathways. The observed classified landscape is therefore temporally stable under the model's transition rules.

04 CONTINUOUS SPECTRAL EVIDENCE

Classification can remain unchanged while vegetation condition shifts within a class. Change Vector Analysis (CVA) therefore measures the magnitude of movement through the NDVI–GNDVI–NDRE feature space.

INDICATOR	MEAN	MEDIAN	P95	WHAT IT REPRESENTS
CVA magnitude	0.6969	0.6719	1.3386	Overall spectral trajectory
ΔNDVI	0.4944	0.2951	1.0850	Photosynthetic vigour / canopy response
ΔGNDVI	0.2689	0.0051	0.8085	Green reflectance / canopy density
ΔNDRE	-0.0334	-0.0046	0.3494	Red-edge / foliar condition





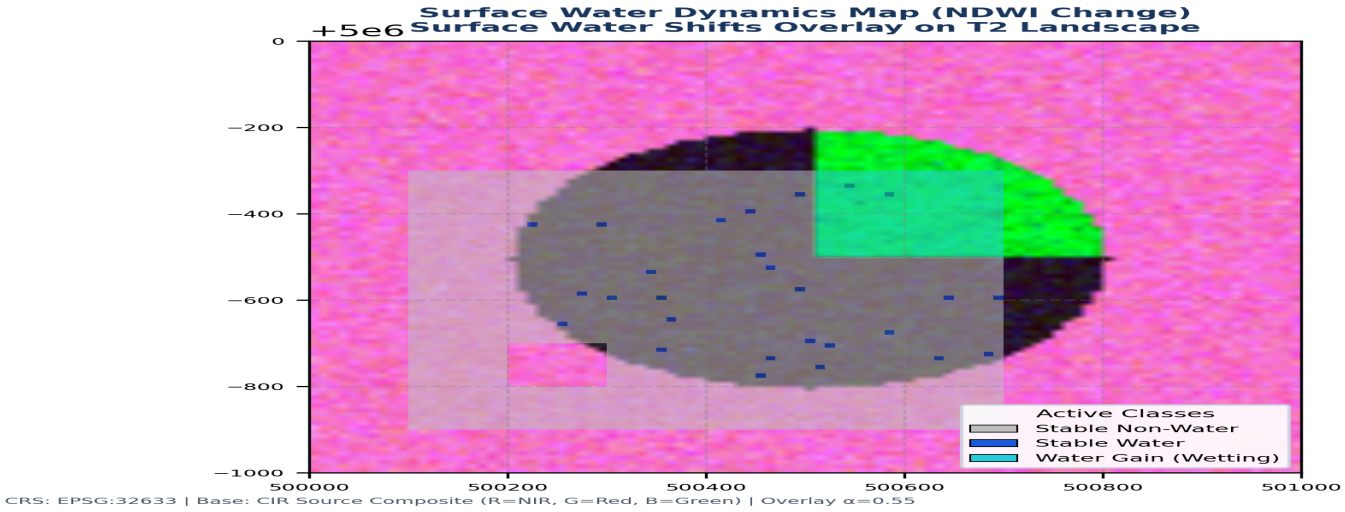
Interpretation

The continuous spectral evidence records measurable displacement. This signal is supporting evidence of vegetation response and is not, by itself, a measure of physical wetland loss.

05 HYDROLOGICAL EVIDENCE

Wetland condition cannot be reduced to vegetation alone. Surface-water behaviour provides a separate hydro-ecological line of evidence. In this engine configuration, water presence is modelled from Green/NIR NDWI using a threshold of 0.00.

OBSERVATION	VALUE	INTERPRETATION
T1 surface-water extent	0.27 ha	Water classified at the baseline date
T2 surface-water extent	4.05 ha	Water classified at the present date
Surface-water gain	3.78 ha	Newly detected water / wetting
Surface-water loss	0.00 ha	Previously detected water becoming non-water / drying
NDWI threshold	0.00	Configured water-classification threshold



Interpretation

Surface-water change reports 0.00 ha of loss and 3.78 ha of gain.

06 FINAL WETLAND ASSESSMENT

The conclusion below is a synthesis of the observations reported in this run. It deliberately distinguishes **vegetation degradation**, **hydro-ecological evidence**, and **physical wetland depletion** because these are not equivalent measurements.

EVIDENCE	OBSERVATION	WHAT IT SUPPORTS
Wetland context	Project-provided wetland boundary	Defines the spatial context in which the ecological signals are interpreted.
Vegetation transitions	0 degradation px; 0 recovery px	Whether classified vegetation condition moved toward deterioration or recovery.
Continuous spectral change	CVA mean 0.6969	Whether average vegetation spectral state shifted even without a class transition.
Hydrological evidence	Gain 3.78 ha; Loss 0.00 ha	Whether surface-water behaviour provides supporting hydro-ecological evidence.

Professional assessment

The T1–T2 comparison contains measurable change in vegetation and/or spectral state within the wetland analysis context. The observed change should be interpreted with the hydrological evidence to determine whether there is a coherent hydro-ecological degradation signal.

The continuous spectral indicators (mean ΔNDVI 0.4944, ΔGNDVI 0.2689, ΔNDRE -0.0334) are used as supporting evidence for within-wetland vegetation change.

Physical wetland depletion is not established by this run. The engine explicitly prevents vegetation decline or optical water change from being presented as physical wetland loss without independent supporting evidence.

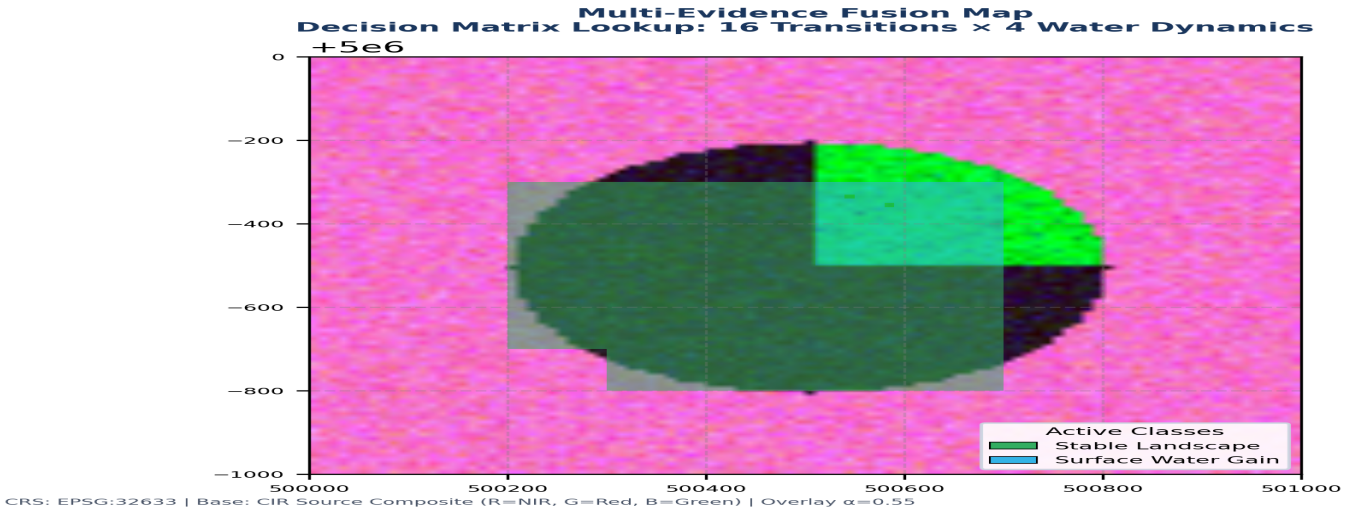
CONCLUSION
NO MEASURABLE VEGETATION DEGRADATION DETECTED

This is a statement about the observed change signal in the analyzed wetland context. It is not a declaration of long-term wetland health or physical wetland persistence.

07 MULTI-EVIDENCE FUSION

The fusion stage combines vegetation transitions with water-state changes. A hydro-ecological degradation signal requires concurrent evidence from both vegetation deterioration and surface-water loss; vegetation-only change is kept separate.

EVIDENCE CLASS	VALUE	MEANING
Stable landscape	3,500 px	Persistent classified state
Hydro-ecological degradation	0.00 ha	Vegetation degradation + surface-water loss
Hydro-ecological recovery	0.00 ha	Vegetation recovery + surface-water gain
Vegetation-only degradation	0 px	Vegetation decline without independent physical depletion confirmation


Multi-Evidence Fusion Map
Decision Matrix Lookup: 16 Transitions × 4 Water Dynamics
+5e6
0
−200
−400
−600
−800
−1000
500000
500200
500400
500600
500800
501000
Active Classes
Stable Landscape
Surface Water Gain
CRS: EPSG:32633 | Base: CIR Source Composite (R=NIR, G=Red, B=Green) | Overlay α=0.55

08 ANALYTICAL QUALITY & BOUNDARIES

The following controls explain how the reported wetland evidence was bounded. They are included so that the ecological conclusion can be traced back to a defined analytical footprint.

CONTROL	STATUS	FINDING
Spatial alignment	PASS	T1 and T2 were accepted on the common analysis grid.
Footprint reconciliation	PASS	10,000 px overlap = 100 px excluded + 3,500 px effective.
Independent ground reference	UNAVAILABLE	No independent reference samples were configured for this run.
Physical depletion guard	ENFORCED	Vegetation or optical water change is not automatically labelled physical wetland loss.

Spatial accounting

FOOTPRINT COMPONENT	PIXELS	PHYSICAL AREA
Original spatial overlap	10,000	100.00 ha
Selected AOI / Inclusion	3,600	36.00 ha
Manual exclusion	100	1.00 ha
Effective analysis footprint	3,500	35.00 ha

When the source raster has no authoritative CRS/GSD, pixel counts are retained as the authoritative spatial measure and hectares are suppressed. This prevents a pixel-space grid from being mistaken for a physically georeferenced wetland area.

09 TECHNICAL PROVENANCE

The report is generated directly from the pipeline result dictionary and the raster artifacts produced during the run. No scientific value is invented by the PDF layer.

INDICATOR	VALUE	UNIT	SOURCE / METHOD
Wetland source	PROJECT_BOUNDARY	class	Wetland mask source selected by wetland context stage
Wetland context status	WETLAND_BOUNDARY_AVAILABLE	status	Wetland mask result
Effective analysis footprint	3,500.00000	pixels	alignment/effective_analysis_mask.tif
Manual excluded	100.00000	pixels	exclusions/manual_exclusion_mask.tif
Vegetation degradation	0.00000	pixels	degradation/degradation_map.tif; H→M, H→S, M→S
Vegetation recovery	0.00000	pixels	degradation/recovery_map.tif; M→H, S→M, S→H
CVA mean magnitude	0.69691	index units	change/cva_magnitude.tif; ||[ΔNDVI, ΔGNDVI, ΔNDRE]||
Mean ΔNDVI	0.49444	index units	change/delta_ndvi.tif; NDVI(T2) – NDVI(T1)
NDWI threshold	0.00000	index units	Green/NIR NDWI threshold
Physical depletion assessment	NOT_ASSESSED	status	Independent-evidence claim guard

Scientific limitations

The T1–T2 interval is short, so seasonal or acquisition-related variation can differ from persistent ecological degradation. Model-derived vegetation classes are inferred outputs and should be field-validated before regulatory or permanent ecological claims are made. Where the wetland context is inferred from optical water, that inference should not be interpreted as a legal wetland delineation.

Run: run_20260922_082307 · **Software:** GIS Engine v1.0.0 · **Area basis:** RASTER_CRS_AND_TRANSFORM